

CUSTOMER SEGMENTATION & RFM ANALYSIS

BUSINESS INSIGHTS REPORT

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Tools Used: Python | Pandas | RFM Analysis | Power BI

1. EXECUTIVE SUMMARY

This Business Insights Report presents the key findings obtained from **Customer Segmentation and RFM (Recency, Frequency, Monetary) Analysis**.

The analysis was performed on customer transaction data using Python. Customers were evaluated based on their purchasing behavior and assigned RFM scores. Based on these scores, customers were divided into five meaningful segments:

- Champions
- Loyal Customers
- Potential Loyalists
- At Risk
- Lost Customers

The final results were visualized using an interactive **Power BI Dashboard**, allowing business users to analyze customer behavior, revenue contribution, and customer value.

Key Finding: The **At Risk** segment is the largest customer segment with **1,188 customers**, indicating a major opportunity for customer retention and re-engagement.

2. CUSTOMER SEGMENTATION OVERVIEW

The RFM analysis resulted in the following customer segments:

Customer Segment	RFM Score	Customers
🏆 Champions	14–15	642
💎 Loyal Customers	12–13	601
🌱 Potential Loyalists	9–11	1,022
⚠️ At Risk	6–8	1,188
😞 Lost Customers	3–5	885
Total	—	4,338

The segmentation provides a clear understanding of the different customer groups and helps businesses design targeted strategies for each segment.

3. KEY BUSINESS INSIGHTS

3.1 🚨 At Risk Customers — Major Retention Opportunity

At Risk customers: 1,188

At Risk is the **largest customer segment** in the analysis.

These customers show weaker engagement compared with the higher-value customer segments. If their engagement continues to decline, they may eventually move into the Lost Customers category.

Business Implication

A significant portion of the customer base requires **retention and re-engagement efforts**.

Recommended Actions

- Launch personalized **win-back campaigns**.
 - Provide targeted discounts and limited-time offers.
 - Send personalized product recommendations.
 - Use email and promotional campaigns to encourage repeat purchases.
 - Monitor customer behavior regularly to identify further decline in engagement.
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3.2 🌱 Potential Loyalists — Growth Opportunity

Potential Loyalists: 1,022

Potential Loyalists represent customers who already show meaningful engagement with the business and have the potential to become long-term loyal customers.

Business Implication

This segment represents a strong opportunity to increase **customer lifetime value and purchase frequency**.

Recommended Actions

- Encourage more frequent purchases.
- Provide personalized product recommendations.
- Introduce loyalty rewards.
- Offer targeted promotions.
- Create personalized campaigns based on previous purchase behavior.

3.3 🧐 Lost Customers — Reactivation Opportunity

Lost Customers: 885

Lost Customers represent customers with low engagement or customers who have not purchased recently.

Business Implication

Some of these customers may still be recoverable through targeted **reactivation and win-back campaigns**.

Recommended Actions

- Launch customer reactivation campaigns.
 - Provide relevant discounts or special offers.
 - Use previous purchase history for personalized communication.
 - Recommend products based on previous interests.
 - Test different offers to identify what brings inactive customers back.
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3.4 🏆 Champions — High-Value Customers

Champions: 642

Champions represent the strongest customer group based on their RFM behavior.

These customers are highly engaged and represent an important source of customer value.

Business Implication

Champions should be protected because losing these customers could negatively affect long-term revenue and customer lifetime value.

Recommended Actions

- Provide exclusive loyalty rewards.
- Offer early access to new products.
- Provide personalized offers.
- Encourage repeat purchases.
- Use cross-selling and upselling strategies.
- Create VIP customer experiences.

3.5 Loyal Customers — Customer Retention

Loyal Customers: 601

Loyal Customers demonstrate strong and consistent purchasing behavior.

Business Implication

Maintaining these customers can help generate **stable and recurring revenue**.

Recommended Actions

- Introduce loyalty programs.
 - Provide personalized recommendations.
 - Encourage cross-selling.
 - Offer exclusive customer benefits.
 - Maintain regular customer engagement.
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4. REVENUE & CUSTOMER VALUE INSIGHTS

The Power BI dashboard provides revenue and average spending analysis across customer segments.

This analysis helps management understand not only **how many customers belong to each segment**, but also **how valuable those customers are to the business**.

Business Importance

Comparing:

Customer Count + Revenue + Average Spending

helps identify which customer segments should receive the highest priority.

Strategic Approach

- High-value customers → **Retention & loyalty**
- Potential customers → **Growth & engagement**
- At-risk customers → **Reactivation & retention**
- Lost customers → **Win-back campaigns**

This creates a more targeted and data-driven marketing strategy.

5. RFM-BASED MARKETING STRATEGY

Customer Segment	Primary Strategy	Business Goal
 Champions	Loyalty & VIP treatment	Retain high-value customers
 Loyal Customers	Loyalty & cross-selling	Increase customer lifetime value
 Potential Loyalists	Engagement & promotions	Convert into loyal customers
 At Risk	Win-back & retention	Prevent customer churn
 Lost Customers	Reactivation	Recover inactive customers

This segmentation allows the business to avoid a **one-size-fits-all marketing approach**.

6. KEY BUSINESS RECOMMENDATIONS

1. Prioritize At Risk Customers

Since At Risk is the largest segment with **1,188 customers**, retention should be one of the major business priorities.

2. Protect High-Value Customers

Champions and Loyal Customers should receive personalized experiences, loyalty benefits, and exclusive offers.

3. Convert Potential Loyalists

Potential Loyalists should be encouraged to purchase more frequently through targeted promotions and personalized recommendations.

4. Reactivate Lost Customers

Lost Customers should be targeted through carefully designed reactivation and win-back campaigns.

5. Personalize Marketing

Customer purchase history and RFM characteristics should be used to create personalized campaigns rather than sending the same offer to every customer.

6. Continuously Monitor Customer Behavior

RFM scores should be monitored regularly so that customers moving from Loyal → At Risk or At Risk → Lost can be identified early

7. EXPECTED BUSINESS IMPACT

Implementing the above strategies can help the business:

-  Increase customer retention
 -  Increase repeat purchases
 -  Improve customer lifetime value
 -  Reduce customer churn
 -  Improve marketing campaign effectiveness
 -  Strengthen relationships with high-value customers
 -  Recover potentially lost customers
 -  Support data-driven decision making
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8. POWER BI DASHBOARD

The interactive Power BI dashboard provides a consolidated view of customer behavior and segment performance.

Dashboard Components

- **Total Customers KPI**
- **Total Revenue KPI**
- **Total Orders KPI**
- **Average RFM Score KPI**
- **Customer Segment Distribution**
- **Customers by Segment**
- **Revenue by Customer Segment**
- **Average Spending by Segment**
- **Recency vs Customer Spending**
- **Frequency vs Customer Spending**

Interactive Filters

- Customer Segment
- RFM Score
- Recency (Days)

The dashboard allows users to interactively filter the customer base and analyze different segments.

9. CONCLUSION

- 1.The RFM-based customer segmentation analysis provides a clear understanding of customer behavior and business value.
 - 2.The most important finding is that **At Risk customers form the largest segment with 1,188 customers**, creating a significant opportunity for customer retention and re-engagement.
 - 3.At the same time, **Champions and Loyal Customers** should be protected through loyalty-focused strategies, while **Potential Loyalists** should be nurtured toward stronger long-term relationships.
 - 4.**Lost Customers** can be targeted through reactivation campaigns designed around their previous purchasing behavior.
 - 5.Overall, the analysis demonstrates how **Python, RFM Analysis, and Power BI** can transform raw transaction data into actionable business insights and support better **customer retention, targeted marketing, and revenue growth**.
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Project: Customer Segmentation & RFM Analysis

Technologies: Python | Pandas | RFM Analysis | Power BI